

Pockets of Predictability: A Replication

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Abstract

Farmer, Schmidt, and Timmermann (FST) document time-variation in market return predictability, identifying “pockets” of significant predictability through kernel regressions. However, our analysis reveals a critical discrepancy between the method outlined by FST and the code actually implemented. Instead of using a one-sided kernel, which guarantees out-of-sample forecasts, they perform in-sample estimation with a two-sided kernel. As a result, future information leaks into the forecasting model, undermining its reliability. Rectifying this error qualitatively alters the findings, invalidating most conclusions of the FST study. Thus, attempts to exploit such “pockets”—should they genuinely exist—offer little help in forecasting market returns.

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1. Introduction

Is it possible to predict aggregate stock market returns? Farmer, Schmidt, and Timmermann (2023, FST) propose a nuanced answer to this long-debated question: *sometimes*. Using nine decades of U.S. data, they identify brief intervals of significant predictability amid longer periods with no return predictability. Crucially, they show that these “pockets of predictability” may be detected *ex ante* through one-sided kernel regression. Consequently, investors can exploit them and generate measurable economic returns through market timing.

In this paper, we spotlight a critical discrepancy between the methodological approach outlined in the study and the code implemented. The forecasting procedure in the paper involves two steps, both based on kernel regression: i) estimation of the parameters of the prediction model, and ii) identification of the pockets of predictability. Notably, the authors claim to employ one-sided kernel regression in both stages. This fundamental approach protects against the inclusion of future information, thereby ensuring a valid out-of-sample approach. As detailed in FST (p. 1288), “all of our estimates are computed recursively, out-of-sample, using only real-time information available prior to the period for which returns are predicted.” However, a review of the actual code included in the replication package reveals that the one-sided kernel regressions are used only in the first step. On the other hand, the second step leverages two-sided kernel regressions, implying an in-sample approach. As a result, the methodology adopted to detect these pockets inadvertently relies on future information, unavailable at the time of the forecast.

To assess the impact of this inconsistency, we replicate all of FST's key analyses using two parallel frameworks. First, we adopt their original replication code, which includes two-sided kernel regression in the second-stage analysis, which is equivalent to in-sample estimation. This method does not prevent information leakage and thus incorporates future data not available during the forecast. Second, we refine the code to adhere to the presented methodology by using one-sided kernel regression in the second stage, which is equivalent to out-of-sample estimation.. This revised approach relies solely on historical data, ensuring that no future information enters the model, thus maintaining its integrity.

The comparison of the results derived from the two frameworks reveals significant differences, effectively debunking the main conclusions of the FST study. One-sided kernel regression proves inefficient at detecting pockets of predictability—assuming they exist. The pockets uncovered using the corrected code are severely truncated, with durations commonly shrinking to no more than one month—a tenfold reduction compared to the original analysis. Furthermore, their frequency escalates to over twenty times that initially documented. The primary concern, however, is the diminished predictability of returns. Figure 1 illustrates a snapshot of these critical findings. The original code generates an average integral R^2 ranging from 1.51% to 3.70% for daily forecasts. In stark contrast, the

revised code records a figure that drops by a factor of about twenty, ranging from just 0.09% to 0.28%.

Figure 1. In-Sample Versus Out-of-Sample Pocket Identification

This figure compares the integral R^2 as a measure of return predictability within pockets identified using two methods: (i) the original code from Farmer et al. (2023), which uses a two-sided kernel regression and corresponds with an in-sample estimation (green bars), and (ii) a replication based on a one-sided, which corresponds with out-of-sample estimation (orange bars). Coefficients are estimated using daily data. The pockets are defined periods when the fitted squared forecast error differential (SED) resulting from a one-year estimation period, comparing the prevailing mean with kernel-based forecasts, exceeds zero in the preceding period. The study period starts on November 5, 1926, for the dividend price ratio, January 4, 1954, for the Treasury bill rate, January 2, 1962, for the term spread, and January 15, 1927, for the realized variance, and always runs through the end of 2016.



Essentially, the results of the tests with the corrected code lead to two principal conclusions. First, there are no substantial differences between the return predictability inside and outside the pockets identified by the one-sided kernel. Second, the return predictability in the in-pocket periods is very weak. Consequently, investors are unlikely to benefit from the alleged pockets through simple market timing strategies.

These findings are supported by extensive tests and robustness checks replicated from FST. After correcting the code, neither univariate forecasting models based on popular variables—dividend yield, T-bill rate, term spread, and realized variance—nor their combinations into composite predictors can effectively predict stock market returns. In particular, the forecasting models fail to significantly outperform a simple prevailing mean benchmark model, both inside and outside the pockets. As a result, attempts to exploit economic gains through pocket-based strategies may prove futile. The market timing techniques proposed by FST show meager abnormal returns that are largely statistically

insignificant. Moreover, the Sharpe ratios derived from these portfolios consistently lag behind those of the naïve model.

Our conclusions are robust to various changes in the underlying methodology proposed by FST. They hold across alternative kernel bandwidth settings, withstand evaluation with multifactor models, and remain qualitatively similar for both daily and monthly data. In all of these tests, the procedure described in FST fails to reliably detect the periods characterized by strong local return predictability.

Finally, despite these predominantly negative results, we identify one positive aspect in our replication. The market timing strategies derived from the kernel models have the potential to be improved by imposing economic constraints on the forecasts. By enforcing positive signs of forecasted returns or coefficients, we find that it is still possible to observe some alphas on market timing strategies, albeit to a lesser extent and with diminished magnitude and significance than the results documented in the FST study. In this way, the models benefit from well-known properties of economic restrictions that eliminate extreme and likely incorrect forecasts—such as unusually negative returns—ultimately improving prediction performance.

The remainder of this article is organized as follows. Section 2 discusses the data and methodology. Section 3 presents the empirical results. Finally, Section 4 concludes the study.

2. Data, Methodology, and Discrepancies

2.1. Data

Our study concentrates on the validity of FST findings using their original data and code. In this regard, our exercise is the closest to the definitions of “pure replication” by Hamermesh (2007), “reexamination” by Welch (2019), or “verification” by Clemens (2017). To avoid semantic confusion, we refer to our procedures simply as replication. In particular, we recreate all key findings within their sample and study period, strictly adhering to their original data, methodology and assessment metrics. This approach ensures an unswerving comparison, grounded on the precise data and codes employed by FST, all sourced from the original replication package labeled “Replication-code 20190881.zip,” available on The Journal of Finance website (<https://onlinelibrary.wiley.com/doi/abs/10.1111/jofi.13229>). The sole deviation implemented in our analysis involves modifying the kernel type designated for pocket identification, a subject elaborated upon in Section 2.3. For an in-depth comprehension of details not covered in our manuscript, readers are encouraged to consult the FST paper and the accompanying replication package.

As in FST, our research focuses on the U.S. stock market. In all our tests, the predicted dependent variable is the excess aggregate market return, calculated as the CRSP U.S. stock market return minus the one-day return on a short T-bill rate. The predictor variables include the dividend-price ratio (dp), three-month T-bill rate (tbl), term spread (tsp), and realized variance ($rvar$). The study period runs from 1926 to 2016, as available for the different series.

2.2. Methodology

FST operate under the assumption that the predictability of market returns varies over time and that it is possible to identify different market stages *ex ante*. They designate the periods during which forecasting is notably effective as “pockets of predictability” and seek to identify them using kernel regressions. To maintain consistency with their work, we adhere to the original notation used by FST in all the equations presented below.

The general return prediction model could be described as:

$$r_{t+1} = x_t' \beta_t + \varepsilon_{t+1}, \quad (1)$$

where r_{t+1} is the excess stock market return at $t+1$, β_t are time-varying regression coefficients, x_t is the vector of predictor variables, and ε_{t+1} denotes an unobservable disturbance term. The model allows for general forms of conditional heteroskedasticity, so that $\sigma_t^2 \equiv \mathbb{E}[\varepsilon_t^2 | x_t] = \sigma^2(x_t)$. The parameters β_t are estimated using a local constant model of the following form:

$$\hat{\beta}_t = \arg \min_{\beta_0 \in R^d} \sum_{s=1}^T K_{hT}(s-t) [r_{t+1} - x_s' \beta_0]^2, \quad (2)$$

where the weights on local observations are determined by the kernel $K_{hT}(u) \equiv K(u/hT)/(hT)$, with h denoting the bandwidth. FST examine whether local predictability could have been identified with a one-sided version of the Epanechnikov kernel:

$$K(u) = \frac{3}{2} (1 - u^2) 1\{-1 < u < 0\}. \quad (3)$$

The term *one-sided* in this context indicates that only the data from before time t , that is, from the past, are considered. Specifically, the authors indicate using 2.5-year bandwidth to perform the calculations. Consequently, this method could be regarded as fully out-of-sample—it does not incorporate any information unknown at time t to form the forecasts.

To determine the fluctuations in predictive accuracy over time, FST calculate the squared error difference, which juxtaposes the prediction errors on of the prevailing mean

benchmark forecast, $r_t - \bar{r}_{t|t-1}$, with the error on the examined kernel regression model, $r_t - \hat{r}_{t|t-1}$:

$$SED_t = (r_t - \bar{r}_{t|t-1})^2 - (r_t - \hat{r}_{t|t-1})^2. \quad (4)$$

Periods of positive SED_t could be interpreted as those when the kernel regression model outperforms the benchmark, generating smaller forecast errors. On the other hand, negative SED_t scores imply less accurate forecasts and kernel model underperformance.

With the SED_t values at hand, FST project them on a constant and a time trend to predict the pockets of predictability, the periods when returns are more predictable:

$$\widehat{SED}_t = \hat{\gamma}_{0,t} + \hat{\gamma}_{1,t}t > 0. \quad (5)$$

As indicated in the referred paper, the parameters $\hat{\gamma}_{0,t}$ and $\hat{\gamma}_{1,t}$ should be, again, estimated with a one-sided Epanechnikov kernel with one-year bandwidth. At the beginning of a pocket, it is anticipated that $\hat{\gamma}_{1,t} > 0$, so recent squared forecast error values from the benchmark model will start to surpass those derived from the local kernel model, signaling the onset of a pocket of predictability. In contrast, once the measure reaches its peak, $\hat{\gamma}_{1,t} < 0$ signals diminishing return predictability.

2.3. Code Discrepancy and the Replication Design

The estimation of the coefficients $\hat{\gamma}_{0,t}$ and $\hat{\gamma}_{1,t}$ in Eq. (5) should be based on a one-sided Epanechnikov kernel regression, which imposes the condition that $-1 < u < 0$. As noticed by FST, this approach guarantees that “all of our estimates are computed recursively, out-of-sample, using only real-time information available prior to the period for which returns are being predicted” (FST, p. 1288). The predictability pockets are then defined as periods (days), when the inequality expressed by Eq. (5) holds, and the forecasts from the local kernel regression should be used. This framework has intuitive, practical inclinations as it corresponds with forecasting future returns without any information unavailable prior to the day of the forecast.

Nevertheless, a closer inspection of the replication code uncovers a significant discrepancy between the methodology outlined in the paper and the actual implementation carried out by FST. The Internet Appendix A presents the technical details, while below, we provide a brief summary of the problem. Contrary to what is described in the research design, the authors use the one-sided kernel only in estimating the parameters of the return prediction model, while the “pockets of predictability” are identified using a two-sided kernel. In other words, the data used to identify pockets encompass both the period preceding and following the forecast date (day t). To elaborate, with a one-year bandwidth, the estimations draw from a whole 24-month period, equally divided to include the 12 months

before and the 12 months after day t . This method aligns with an in-sample estimation, not an out-of-sample approach as originally claimed. To gauge the ramifications of this methodological discrepancy, we replicate all the key results by using two distinct approaches:

In-sample pocket identification (two-sided kernel). This approach reproduces the outcomes using the exact code available from the replication package despite its deviation from the framework outlined in the paper. It adopts a two-sided kernel ($-1 < u < 1$), consistent with in-sample pocket identification, leveraging information unavailable at the time t .

Out-of-sample pocket identification (one-sided kernel). This method rectifies the discrepancy in the original approach by adhering to the methodology outlined in the paper. It employs a one-sided kernel ($-1 < u < 0$), as delineated in Eq. (3), aligned with out-of-sample pocket identification, and restricts the analysis to data available before time t .

The details on how the code is implemented and corrected in both versions are available from the Internet Appendix A. Aside from the modification above to the kernel function, no further changes were made to the implementation code. Importantly, we maintain the one-year bandwidth stipulation, effectively utilizing 12 and 24 months of data in the estimation window for the one-sided and two-sided kernels, respectively.

3. Empirical Findings

We begin the discussion of the results with an initial evaluation of the pocket characteristics. Next, we continue with a more formal assessment of their statistical and economic significance, followed by additional robustness checks and further insights.

3.1. Initial Evaluation

We begin with an overall evaluation of the market return predictability using daily data and univariate regressions. Table 1 reports the replication results for the constant coefficient regressions to predict daily stock market returns with dp , tbl , tsp , and $rvar$.¹ Panel A concentrates on the in-sample pocket identification method, reproducing the FST results using their original unchanged code. Overall, the numbers are identical to FST, confirming that the authors' code properly recreates the original findings. Not surprisingly, all the major FST conclusions hold: the return predictability is weak outside the pockets but thrives within them.

¹ Note that there is no pocket identification for the full sample. For this reason, the results for the full sample in the tables do not differ when we distinguish between in-sample and out-of-sample.

[Insert Table 1 here]

The in-sample $\bar{R}^2$ values for the full study period (table's top section) are relatively low, ranging from virtually 0% for realized variance to 0.053% for Treasury bill yields. The corresponding slope coefficients are significant only for T-bill rates and term spread. Focusing only on the in-pocket periods (the middle section) marks a qualitative change. Both slope coefficients and t -statistics visibly swell in absolute terms, with the dividend-price ratio also joining the club of significant predictors. The $\bar{R}^2$ figures for those variables, range from 0.179% (dp) to 1.466% (tsp), marking a substantial increase. On the other hand, the return predictability in the out-of-pocket periods is non-existent. None of the slope coefficients is significant, and the $\bar{R}^2$ values practically amount to zero. In summary, the findings seemingly confirm a substantial and foreseeable time variation in return predictability.

Observe now the results for the out-of-sample estimations, presented in Table 1, Panel B. The calculations based on the one-sided kernel visibly contrast those in Panel A. The significant differences between in-pocket and out-of-pocket periods largely disappear. The $\bar{R}^2$ scores are comparably low in both market regimes, and only the term spread continues to exhibit a significant slope coefficient in the out-of-sample period. The relevant in-pocket $\bar{R}^2$ scores span from -0.011% for the dividend-price ratio to 0.376% for the term spread. Noteworthy, the out-of-sample approach even detects out-of-pocket predictability for T-bill rates.

Table 2 sheds further light on pocket characteristics. Following FST, we count how many pockets exist within the sample, measuring also how long they last. The duration of a single pocket is defined as the number of consecutive observations classified as within the pocket. Overall, the more extended pockets benefit investors, as they can be more easily uncovered and exploited. Furthermore, we calculate R^2 coefficients and the integral R^2 (IR^2) to capture the magnitude of predictability within a pocket (for details, see page 1289 in FST).

[Insert Table 2 here]

Notably, as in FST, we display the outcomes for daily and monthly data. Again, the results of in-sample estimation (Panel A) perfectly match the numbers in the original study by Farmer et al. (2023). We observe a relatively modest number of pockets—between 7 and 18—with a considerable duration. The pockets last, on average, between 149 and 303 days, depending on the specifications. Furthermore, in certain cases, they even run for multiple years, and their length hardly ever falls below one month. Finally, the return predictability within the pockets is notable. The average integral R^2 values range from 1.48% to 3.70%; in some pockets, they even exceed 16%. In summary, the investors may benefit from long-lived periods of superior predictability, which could be exploited for profits.

Contrary to the optimistic perspective in Panel A, our code correction embracing the out-of-sample approach leads to a remarkably different picture (Table 2, Panel B). The pocket characteristics change fundamentally. They become much more prevalent but substantially shorter, with many of them lasting just for one trading day. Specifically, they occur roughly 20-30 times more often and are more than ten times shorter. For example, for the daily data, we detect between 252 and 431 pockets of the average duration ranging between 16 and 26 days. In total, the pockets cover from 33% to 43% of the entire study period. Nonetheless, the return predictability within them is nowhere close to that in Panel A. For the daily data, the average integral R^2 scores span from 0.09% to 0.28%, and it increases only slightly for the monthly data to reach the range of 0.42% to 0.63%.

To conclude, accurately accounting for the out-of-sample forecasts in return predictability leads to qualitative changes in pocket characteristics. Employing a one-sided kernel instead of a two-sided kernel renders the forecasts more frequent yet shorter, significantly reducing their ability to inform future stock market returns.

3.2. Statistical and Economic Performance of Return Forecasts

Having assessed the baseline properties of the in-sample and out-of-sample estimations, we now move to more rigorous inferences. Specifically, we extend our analyses in three dimensions: (i) predictor variables, (ii) restrictions on forecasts, and (iii) performance measurement.

3.2.1. Tests and Performance Measurement

In terms of predictor variables, our analyses thus far relied on well-established individual predictors. Since multiple predictors may contain independent incremental information, we now broaden our tests to five composite measures: a recursively computed first principal component of the four individual variables (*pc*), a four-variable multivariate forecast estimated using a product kernel (*mv*), and three distinct averages of univariate forecasts, denoted *comb1*, *comb2*, and *comb3*. *comb1* assigns the forecast from the time-varying coefficient model in-pocket but defaults to the prevailing mean out-of-pocket. It then takes the average from all models. *comb2* takes the average of all models indicating a pocket and the prevailing mean if no model indicates a pocket. *comb3* consistently utilizes the equal-weighted mean of the four standalone models, regardless of whether in-pocket or out-of-pocket. Further details of all the composite measures are available from FST. Regarding the economic constraints, as in Campbell and Thompson (2008), we account for three ways of incorporating restrictions into the forecasts: (i) unrestricted forecasts, $\hat{r}_{t+1|t}$; (ii) restrictions requiring nonnegative forecasts that replace negative predictions with zero, $\max(0, \hat{r}_{t+1|t})$; and (iii) restrictions on slope coefficients that require both nonnegative forecasts and the slope coefficient to be consistent with prior

expectations concerning the sign (negative for *tbl*, positive for *dp*, *tsp*, *rvar*, and composite measures). The last constraint sets $\hat{\beta}_t = 0$ for the inconsistent signs.

Finally, concerning the performance measurement, the examination involves several additional popular metrics. First, we calculate the Clark and West (2007, CW) test, which compares the predictive accuracy of the kernel-based models with a standard benchmark—the prevailing mean model, $\bar{r}_{t+1|t} = \frac{1}{t} \sum_{s=1}^t r_s$ (see, e.g., Welch & Goyal, 2008). Second, to assess the economic significance of the forecasts, we test an asset allocation strategy that shifts funds between the stock market and Treasury bills depending on the return expectations, $E_t[r_{t+1}]$. For the details on the implementation of this strategy, please refer to pages 1297-98 in FST. Importantly, following the convention of FST, the statistical significance of alphas is assessed using one-tailed hypothesis tests.

3.2.2. Results

Table 3 reproduces the FST analysis of a measure of prediction performance using both one-sided and two-sided kernels. Panel A presents the CW *t*-statistics, with Panels A.1 and A.2 representing the in-sample and out-of-sample pocket identification methods. The in-sample tests mirror the inferences drawn in the FST analysis closely, highlighting a pronounced discrepancy between the return predictabilities observed during in-pocket and out-of-pocket periods. For all three approaches to economic restrictions, nearly all CW tests indicate strong outperformance of the kernel-based models over the prevailing mean benchmark forecast. The in-pocket CW *t*-statistics are remarkably high, commonly exceeding the threshold of 3 or even 4. In contrast, during the out-of-pocket periods, the CW statistics typically fall below zero. In roughly half of the cases, their values are significantly negative (though only at the 10% level), signaling that the prevailing mean benchmark model beats the more sophisticated methods.

[Insert Table 3 here]

However impressive the outcomes in Panel A may look, they still suffer from a critical flaw: the reliance on future information unavailable at the time of forecast. Amending this issue by using the one-sided kernel qualitatively alters the findings. Table 3, Panel A.2 illustrates the results of the out-of-sample pocket identification method, which employs a one-sided kernel and relies only on information available before the forecast time. Not only do we detect no substantial differences between the in-pocket and out-of-pocket periods, but also hardly any in-pocket predictability is observable at all.

The CW *t*-statistics turn out insignificant for most cases—in pockets and outside them alike—implying that the kernel-based prediction models fail to beat the default forecasts based on the long-term mean. There are only a few exceptions, mainly concerning *pc*, *comb1*, and *comb2* measures, which seem to perform well both in the full sample and in

the in-pocket periods. Nevertheless, even in these cases, the in-pocket is typically comparable to the full sample predictability. A comprehensive analysis of the 27 configurations examined in Table 3 reveals a scant number of scenarios where the benchmark model is outdone—seven instances in total sample evaluations and nine within in-pocket periods. This data does not yield convincing evidence to endorse the superiority of return predictability during in-pocket intervals, thereby casting doubt on the central thesis advocated in the original study.

Table 3, Panel B, adopts a more practical perspective on the forecast effectiveness by examining their economic benefits from an application to an asset allocation strategy. Generally, the findings corroborate the results from Panel A: There is a qualitative difference between the in-sample and out-of-sample results.

Focusing initially on the straightforward approach—the forecasts with no additional restrictions, reported in the furthest left section—we observe that asset allocation strategies grounded on the two-sided kernel for pinpointing pockets of predictability (as seen in Panel B.1) yield remarkable performance. The generated alphas range from 0.76% to 6.38%, showcasing statistical significance in all scenarios. Sharpe ratios oscillate between 0.43 and 1, averaging at 0.71 across the considered prediction models.

Contrastingly, adopting more realistic assumptions regarding data availability by utilizing a one-sided kernel (detailed in Panel B.2) leads to a pronounced decline in performance. Alphas drop to the range of -0.44% to 2.51%, with only four instances of significant departures from zero, including only one surpassing the 1% significance threshold. The Sharpe ratios vary from 0.38 to 0.57, averaging at a diminished 0.44, typically falling beneath the level achieved by the prevailing mean benchmark model (0.46). In summary, when the models rely solely on historical data, their practical benefits—if any—are minimal.

Panel B's middle and right sections pertain to forecasts incorporating additional economic constraints. Intriguingly, imposing restrictions on the sign of the forecast or coefficients improves the model performance, boosting both alphas and Sharpe ratios. Take, for example, the restrictions on the sign of the predicted return documented in the central section. Here, the average annualized alpha for the out-of-sample methodology (observed in Panel B.2) stands at 2.3%, in contrast to the 5.1% noted in the in-sample framework (delineated in Panel B.1). In other words, although we still observe a substantial decline in abnormal returns, they remain positive and significant in one-tailed hypothesis tests for most models. In a similar vein, the average Sharpe ratio declines from 0.81 in Panel B.1 to 0.49 in Panel B.2. Again, although the decrease is considerable, the new models still marginally beat the prevailing mean benchmark model, which exhibits the Sharpe ratio of 0.46.

FST observe a similar phenomenon in their seminal paper, where restricted models notably outperformed their unrestricted counterparts. This reflects the well-understood benefits of economic constraints, which shrink the forecasts thereby improving the model's predictive performance. Campbell and Thompson (2008), who were the first to propose the discussed restrictions, noted that curbs on the signs of the coefficients and forecasts consistently improve the prediction performance. Specifically, they allow for eliminating extreme or likely incorrect forecasts, such as exceptionally negative returns. Moreover, the restricted models particularly excelled in times characterized by formidable forecasting challenges, engendering substantial utility gains even with variables possessing feeble predictive capabilities. Campbell and Thompson (2008, p. 1528) interpret this by evoking the well-known econometric principle that “even false theoretical restrictions can be helpful in forecasting if they reduce the variance of a predictor more than they increase its bias.”

3.3. Alternative Bandwidths

FST specify that their baseline approach assumes a bandwidth of 2.5 years in the local kernel regression to forecast returns and a one-year bandwidth to identify the pockets of predictability. Noteworthy, since the original code utilizes a two-sided instead of a one-sided kernel in the latter approach, the actual estimation window in pocket identification equals two years.

To ascertain robustness to the bandwidth choices, FST experiment with alternative lengths. First, they indicate considering 2-, 2.5-, and 3-year windows for the parameter estimation. Second, they run pocket predictions with 6-, 12-, and 18-month bandwidths (effectively, representing 12-, 24-, and 36-month estimation windows). The analysis encompasses all nine individual predictor variables discussed in Section 2.1, as well as both unrestricted and restricted return forecasts.

We recreate this analysis, and Table 4 tabulates the outcomes. The alternative bandwidth choices substantiate all our earlier conclusions. While effectively the results in Panel A are based on the in-sample pocket identification with the two-sided kernel utilized in the original code, they demonstrate consistently impressive performance. The return predictability is impressively solid in the in-pocket periods and practically non-existent outside the pockets. This pattern holds for practically all predictors and forecasting variants. Nevertheless, modifying the code to exclusively utilize the one-sided kernel for pocket identification fundamentally alters the outcomes (Panel B). The predictability completely disappears from the in-pocket periods; not a single instance of significant CW values is recorded. In other words, the kernel-based prediction models fail to discern the periods of superior return predictability.

[Insert Table 4 here]

In Table B1 in the Internet Appendix B, we supplement the findings above with the analysis of restricted forecasts. Admittedly, this operation leads to partial improvement in the results. We do observe a certain number of significant values from the CW test—especially for the composite predictions. Nevertheless, they still represent a minority, and the results are substantially weaker than the original findings based on a two-sided kernel. Finally, as already noticed, all conclusions from the constraint forecasts should be considered with caution as they may result from the restriction mechanics rather than the forecasting model quality.

3.4. Monthly Data

Since our analyses thus far relied mainly on the daily data, for robustness, we now reproduce Table 3 using monthly returns. More specifically, we closely replicate the tests from Section H in FST—using both one- and two-sided kernels for pocket identification.

As seen in Table 5, we observe no qualitative change to our baseline findings. As before, while the two-sided kernel in Panel A generates impressive performance in the in-pocket period and fails to outperform the benchmark model in the out-of-pocket regimes, no similar variability is visible for the one-sided kernel (Panel B). In essence, the forecasting models generate qualitatively similar performance in within and outside the pockets, and the few scattered significant CW tests cluster mainly in the restricted forecasts. Supplementary findings in Table B2 in the Internet Appendix, which focuses on economic significance, lead to similar conclusions. In summary, once the proper out-of-sample estimation is used, the superior return predictability in the pockets either visibly weakens or entirely disappears.

[Insert Table 5 here]

3.5. Further Robustness Checks and Additional Insights

In our final tests, we extend our analyses in two directions. First, we include additional factors in the calculation of alpha for the market timing strategies, effectively replicating the analyses in Table VI in the FST. The results of these tests consistently support our basic findings: correcting for the error in FST invalidates most of the abnormal return results. For brevity, we report these tests in detail in the Internet Appendix C to this paper.

Second, we examine the local predictability of the size and value factors, focusing mainly on small-minus-big (SMB) and high-minus-low (HML) portfolios. Specifically, we apply the FST approach to investigate pockets of predictability in factor returns.

For brevity, we report the details of these analyses in the Internet Appendix D. Interestingly, unlike aggregate stock returns, factor returns retain some of the time-

varying patterns documented in FST. Specifically, we observe significant differences in return predictability between the in- and out-of-pocket periods, as well as mostly significant CW test statistics. Moreover, these patterns translate into measurable gains from market timing strategies. Admittedly, the overall effect is weaker and less prevalent than documented by FST using two-sided kernel regression—especially for HML—but it remains generally significant.

4. Conclusion

The paper by FST argued that the stock market returns exhibit “pockets of predictability,” which could be identified ex-ante using kernel regressions. However, contrary to the described methodology, they employ in-sample estimation with two-sided kernels when identifying pockets of predictability rather than one-sided kernels synonymous with an out-of-sample approach. As a result, future information—unavailable at the time of the forecast—leaks into the model, undermining its reliability. Correcting this error changes the findings qualitatively, invalidating most of the FST findings. Consequently, attempts to leverage such identified “pockets,” assuming they exist, may prove unfruitful.

References

- Campbell, J. Y., & Thompson, S. B. (2008). Predicting excess stock returns out of sample: Can anything beat the historical average? *Review of Financial Studies*, 21(4), 1509-1531.
- Clark, T. E., & West, K. D. (2007). Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics*, 138(1), 291-311.
- Clemens, M. A. (2017). The meaning of failed replications: A review and proposal. *Journal of Economic Surveys*, 31(1), 326-342.
- Farmer, L. E., Schmidt, L., & Timmermann, A. (2023). Pockets of predictability. *Journal of Finance*, 78(3), 1279-1341.
- Hamermesh, D. S. (2007). Replication in economics. *Canadian Journal of Economics/Revue canadienne d'économie*, 40(3), 715-733.
- Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703.
- Welch, I. (2019). Reproducing, extending, updating, replicating, reexamining, and reconciling. *Critical Finance Review*, 8(1-2), 301-304.
- Welch, I., & Goyal, A. (2008). A comprehensive look at the empirical performance of equity premium prediction. *Review of Financial Studies*, 21(4), 1455-1508.

Table 1. Replication Results for the Constant-Coefficient Regressions (FST, Table I)

This table presents the performance of prediction models based on the original code from Farmer et al. (2023), which uses a two-sided kernel regression model to identify pockets of predictability (Panel A) and a replication based on a one-sided kernel (Panel B). We report slope coefficients, Newey-West (1987) adjusted t -statistics, and $\bar{R}^2$ coefficients for univariate regressions of daily excess stock market returns on the lagged predictor variables indicated in the first column. The study period starts on November 5, 1926, for the dividend price ratio (dp), January 4, 1954, for the Treasury bill rate (tbl), January 2, 1962, for the term spread (tsp), and January 15, 1927, for the realized variance ($rvar$), and always runs through the end of 2016. The rightmost column of each panel indicates the number of available observations ($\# obs.$).

Variables	Panel A: In-sample pocket identification (two-sided kernel)				Panel B: Out-of-sample pocket identification (one-sided kernel)			
	Slope coefficient	t -statistic	$\bar{R}^2$ (in %)	# obs.	Slope coefficient	t -statistic	$\bar{R}^2$ (in %)	# obs.
<i>Full sample</i>								
dp	0.025	1.142	0.005	23,786	0.025	1.142	0.005	23,786
tbl	-0.007	-2.783	0.053	15,860	-0.007	-2.783	0.053	15,860
tsp	0.017	2.311	0.041	13,846	0.017	2.311	0.041	13,846
rvar	0.000	0.536	0.000	23,727	0.000	0.536	0.000	23,727
<i>In-pocket</i>								
dp	0.084	2.546	0.179	3,483	0.002	0.087	-0.011	9,162
tbl	-0.014	-3.293	0.374	3,506	-0.005	-1.053	0.004	5,146
tsp	0.073	3.953	1.466	1,810	0.050	3.485	0.376	4,083
rvar	0.000	0.136	-0.019	4,841	0.000	1.117	0.060	9,627
<i>Out-of-pocket</i>								
dp	0.012	0.441	-0.004	18,943	0.043	1.393	0.016	13,264
tbl	-0.003	-0.874	-0.002	10,994	-0.007	-1.936	0.040	9,354
tsp	0.006	0.751	-0.005	10,676	0.001	0.090	-0.012	8,403
rvar	0.000	0.656	0.004	17,526	0.000	0.054	-0.008	12,740

Table 2. Replication Results for Pocket Statistics (FST, Table II)

This table presents the statistics on the pockets of predictability identified using two methodologies: the original code from Farmer et al. (2023), which uses a two-sided kernel regression model to identify pockets of predictability (Panel A), and replication based on a one-sided kernel (Panel B). We report the number of pockets, associated sample fraction, duration (in days), and the integral R^2 . Coefficients are estimated using daily or monthly data (left and right sections, respectively). The study period starts on November 5, 1926, for the dividend price ratio (*dp*), January 4, 1954, for the Treasury bill rate (*tbl*), January 2, 1962, for the term spread (*tsp*), and January 15, 1927, for the realized variance (*rvar*), and always ends in 2016.

Statistics	Daily data				Monthly data			
	dp	tbl	tsp	rvar	dp	tbl	tsp	rvar
<i>Panel A: In-sample pocket identification (two-sided kernel)</i>								
No. of pockets	18	12	7	16	15	15	10	18
Fraction of sample	0.16	0.24	0.15	0.22	0.17	0.21	0.12	0.19
Duration								
Min	16	57	95	25	21	21	42	21
Mean	193.5	292.2	258.6	302.6	243.6	207.2	149.1	226.3
Max	610	672	501	1,302	714	588	378	735
Integral R^2								
Min	-0.24	-0.24	0.28	-0.87	0.04	0.06	0.22	-0.29
Mean	1.51	3.70	2.92	2.77	1.79	2.21	1.59	1.48
Max	4.76	11.69	7.54	16.42	6.27	7.43	5.34	5.84
<i>Panel B: Out-of-sample pocket identification (one-sided kernel)</i>								
No. of pockets	431	283	252	375	96	70	70	109
Fraction of sample	0.41	0.35	0.33	0.43	0.40	0.36	0.33	0.42
Duration								
Min	1	1	1	1	21	21	21	21
Mean	21.3	18.2	16.2	25.7	89.5	74.4	59.7	82.1
Max	208	164	271	227	504	273	231	273
Integral R^2								
Min	-0.08	-2.50	-0.40	-0.10	-0.13	-0.51	-0.62	-0.26
Mean	0.18	0.12	0.09	0.28	0.63	0.59	0.42	0.55
Max	2.42	4.08	3.16	11.46	4.62	3.94	3.61	3.13

Table 3. Replication Results for Measures of Prediction Performance (FST, Table III)

This table compares the prediction performance of models based on the original code from Farmer et al. (2023), which uses a two-sided kernel regression model to identify pockets of predictability (in-sample approach, Panels A.1 and B.1) and replication based on a one-sided kernel (out-of-sample approach, Panels A.2 and B.2). Panel A presents the Clark and West (2007) test statistics (CW) for return predictability benchmarked against a prevailing mean forecast. The calculations assume daily benchmark specification. The forecast variables include individual predictors: dividend price ratio (dp), Treasury bill rate (tbl), term spread (tsp), and realized variance ($rvar$), as well as composite measures: a recursively computed first principal component of the four individual variables (pc), a four-variable multivariate forecast estimated using a product kernel (mv), and simple averages of univariate forecasts ($comb1$, $comb2$, $comb3$). $comb1$ assigns the forecast from the time-varying coefficient model in-pocket but defaults to the prevailing mean out-of-pocket. $comb2$ takes the average of all models indicating a pocket and the prevailing mean if no model indicates a pocket. $comb3$ consistently utilizes the equal-weighted mean of the four standalone models, regardless of whether in-pocket or out-of-pocket. Finally, pm refers to the prevailing mean forecast. Positive CW values suggest that the return forecasts outperform the prevalent mean forecast, whereas negative figures signify the opposite. The pockets are defined periods when the fitted squared forecast error differential (SED) calculated using a one-year bandwidth, comparing the prevailing mean with kernel-based forecasts, exceeds zero in the preceding period. Panel B displays three measures of economic significance tied to the returns on a portfolio utilizing the time-varying coefficient model forecast while in-pocket and the prevailing forecast while out-of-pocket for asset allocation between the risk-free instrument (Treasury bills) and the market portfolio: the annualized alpha ($\hat{\alpha}$), the associated Newey and West (1987) adjusted t-statistic ($t_{\hat{\alpha}}$), and the annualized Sharpe ratio. We calculate three variants of the forecasts: unrestricted (left section), with the restriction that forecasts of mean excess returns should be non-negative (middle section), and with the sign restriction on slope coefficients (right section). The asterisks *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels from a hypothesis test of $\beta > 0$, and † , ‡ , and $^{\text{+++}}$ indicate statistical significance at the 10%, 5%, and 1% levels from a hypothesis test of $\beta < 0$. The asterisks for alphas in Panel B are based on one-tailed hypothesis tests as well.

Panel A: Clark and West (2007) t -statistics

Variables	Unrestricted			+ Excess return forecast			All sign restrictions		
	Full sample	In-pocket	Out-of-pocket	Full sample	In-pocket	Out-of-pocket	Full sample	In-pocket	Out-of-pocket
<i>Panel A.1: In-sample pocket identification (two-sided kernel)</i>									
dp	-0.74	3.00***	-1.62 [†]	0.40	3.79***	-1.94 ^{††}	0.68	4.03***	-1.84 ^{††}
tbl	0.68	3.28***	-1.58 [†]	1.98**	4.75***	-1.33 [†]	2.03**	4.69***	-1.21
tsp	0.15	3.04***	-1.52 [†]	0.95	4.52***	-1.54 [†]	0.79	4.18***	-0.21
rvar	-1.49 [†]	2.88***	-1.77 ^{††}	-0.79	3.93***	-1.07	-0.44	3.10***	-0.97
mv	-0.99	3.74***	-1.49 [†]	-0.01	4.01***	-1.22	-0.01	4.01***	-1.22
pc	0.99	2.71***	-0.56	1.85**	4.69***	-0.52	1.85**	4.69***	-0.52
comb1	4.48***	4.58***	–	6.35***	6.55***	–	6.20***	6.39***	–
comb2	4.95***	5.05***	–	6.15***	6.33***	–	6.28***	6.48***	–
comb3	-1.03	2.11**	-2.02 ^{††}	0.23	0.82	-1.23	0.66	2.55***	-1.03
<i>Panel A.2: Out-of-sample pocket identification (one-sided kernel)</i>									
dp	-0.74	-0.47	-0.59	0.40	-0.64	0.87	0.68	-0.17	0.89
tbl	0.68	0.10	0.79	1.98**	2.02**	0.52	2.03**	2.04**	0.54
tsp	0.15	-1.06	1.49*	0.95	0.90	0.44	0.79	1.00	0.43
rvar	-1.49 [†]	-0.77	-1.18	-0.79	0.38	-0.93	-0.44	1.08	-0.79
mv	-0.99	0.03	-1.28	-0.01	0.23	-0.29	-0.01	0.23	-0.29
pc	0.99	0.58	0.71	1.85**	2.19**	0.52	1.85**	2.19**	0.52
comb1	-0.78	-0.70	–	1.12	1.12	–	1.62*	1.63*	–
comb2	-1.40 [†]	-1.25	–	1.11	1.11	–	2.11**	2.14**	–
comb3	-1.03	-1.20	1.07	0.23	0.02	0.68	0.66	0.66	0.17

Panel B: Economic significance

Var.	Unrestricted			+ Excess return forecast			All sign restrictions		
	$\hat{\alpha}$	$t_{\hat{\alpha}}$	Sharpe ratio	$\hat{\alpha}$	$t_{\hat{\alpha}}$	Sharpe ratio	$\hat{\alpha}$	$t_{\hat{\alpha}}$	Sharpe ratio
<i>Panel B.1: In-sample pocket identification (two-sided kernel)</i>									
dp	1.69**	2.09	0.47	2.50***	2.89	0.54	2.95***	3.26	0.57
tbl	3.57***	4.35	0.79	6.47***	5.56	0.94	6.06***	5.36	0.92
tsp	3.14***	4.26	0.77	5.69***	4.95	0.85	5.03***	4.45	0.84
rvar	2.31***	3.62	0.68	2.88***	3.46	0.71	2.69***	3.03	0.54
mv	2.59***	3.37	0.64	4.79***	4.96	0.85	4.79***	4.96	0.85
pc	3.42***	3.96	0.73	5.86***	5.00	0.86	5.86***	5.00	0.86
comb1	6.38***	6.11	1.00	6.71***	6.70	1.05	6.69***	6.45	0.98
comb2	6.10***	5.66	0.87	8.53***	6.69	0.99	8.35***	6.51	0.95
comb3	0.76*	1.32	0.43	2.35**	1.88	0.46	2.73**	2.08	0.48
pm	-0.26 [†]	-1.62	0.46	-0.26 [†]	-1.62	0.46	-0.26 [†]	-1.62	0.46
<i>Panel B.2: Out-of-sample pocket identification (one-sided kernel)</i>									
dp	-0.44	-0.71	0.39	-0.79	-0.67	0.39	-0.35	-0.29	0.38
tbl	1.72**	2.12	0.48	3.68***	2.97	0.58	3.49***	2.82	0.57
tsp	0.71	0.95	0.40	2.65**	2.04	0.48	3.49***	2.85	0.56
rvar	-0.11	-0.24	0.38	1.01	0.95	0.40	1.83**	1.68	0.44
mv	0.93**	2.18	0.49	2.28***	2.69	0.57	2.28***	2.69	0.57
pc	2.51***	3.21	0.57	3.85***	3.04	0.57	3.85***	3.04	0.57
comb1	0.47	0.85	0.40	2.39**	1.98	0.47	3.04***	2.56	0.53
comb2	0.41	0.55	0.39	2.83**	1.92	0.47	4.03***	2.80	0.54
comb3	0.76*	1.32	0.43	2.35**	1.88	0.46	2.73**	2.08	0.48
pm	-0.26 [†]	-1.62	0.46	-0.26 [†]	-1.62	0.46	-0.26 [†]	-1.62	0.46

Table 4. Replication Tests for Robustness of Forecasting Performance (FST, Table V, Panel A)

This table reports the Clark and West (2007) test statistics (CW) for return predictability relative to a prevailing mean forecast, where positive CW values suggest that the model return forecasts outperform those of the prevalent mean forecast, whereas negative figures signify the opposite. The forecasts are based on different combinations of bandwidths for both the coefficient from the predictive regression and fitted squared forecast error differential estimation, where the first number in the head of each column refers to the sample size for coefficient estimation (2 to 3 years), and the second concerns the fitted squared forecast error differential (6 months to 1.5 years). The leftmost column presents the results for the full sample estimation. The calculations are based either on the original code from Farmer et al. (2023), which uses a two-sided kernel regression model to identify pockets of predictability (in-sample approach, Panel A), or a replication based on a one-sided kernel (out-of-sample approach, Panels B). The forecast variables include individual predictors: dividend price ratio (*dp*), Treasury bill rate (*tbl*), term spread (*tsp*), and realized variance (*rvar*), as well as composite measures: a recursively computed first principal component of the four individual variables (*pc*), a four-variable multivariate forecast estimated using a product kernel (*mv*) and simple averages of univariate forecasts (*comb1*, *comb2*, *comb3*). *comb1* assigns the forecast from the time-varying coefficient model in-pocket but defaults to the prevailing mean out-of-pocket. *comb2* takes the average of all models indicating a pocket and the prevailing mean if no model indicates a pocket. *comb3* consistently utilizes the equal-weighted mean of the four standalone models, regardless of whether in-pocket or out-of-pocket. The pockets are defined periods when the fitted squared forecast error differential (SED) calculated using a one-year bandwidth, comparing the prevailing mean with kernel-based forecasts, exceeds zero in the preceding period. The predictions come from unrestricted forecasting models. The asterisks *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels from a hypothesis test of $\beta > 0$, and [†], ^{††}, and ^{†††} indicate statistical significance at the 10%, 5%, and 1% levels from a hypothesis test of $\beta < 0$.

Variables	Full Sample	In-pocket					Out-of-pocket				
		2.5yCoef,	2yCoef,	3yCoef,	2.5yCoef,	2.5yCoef,	2.5yCoef,	2yCoef,	3yCoef,	2.5yCoef,	2.5yCoef,
		1ySED	1ySED	1ySED	6mSED	1.5ySED	1ySED	1ySED	1ySED	6mSED	1.5ySED
Panel A: In-sample pocket identification (two-sided kernel)											
dp	-0.74	3.00***	2.92***	3.43***	3.59***	2.57***	-1.62 [†]	-2.12 ^{††}	-1.62 [†]	-2.28 ^{††}	-1.47 [†]
tbl	0.68	3.28***	3.54***	3.32***	3.88***	3.88***	-1.58 [†]	-1.00	-1.73 ^{††}	-1.61 [†]	-1.98 ^{††}
tsp	0.15	3.04***	2.83***	3.29***	4.17***	2.77***	-1.52 [†]	-1.08	-1.63 [†]	-1.84 ^{††}	-1.38 [†]
rvar	-1.49 [†]	2.88***	2.42***	2.88***	4.60***	1.93**	-1.77 ^{††}	-1.82 ^{††}	-1.83 ^{††}	-2.05 ^{††}	-1.77 ^{††}
mv	-0.99	3.74***	2.65***	4.08***	4.58***	1.68**	-1.49 [†]	-1.41 [†]	-1.29 [†]	-1.80 ^{††}	-1.28
pc	0.99	2.71***	3.36***	2.96***	5.05***	3.55***	-0.56	-0.80	-0.56	-2.08 ^{††}	-1.28
comb1	4.48***	4.58***	4.34***	4.74***	5.27***	4.64***	—	—	—	—	—
comb2	4.94***	5.05***	4.88***	5.25***	5.72***	5.13***	—	—	—	—	—
comb3	-1.03	2.11**	2.68***	2.05**	2.44***	2.31**	-2.02 ^{††}	-2.34 ^{†††}	-1.95 ^{††}	-2.20 ^{††}	-2.11 ^{††}
Panel B: Out-of-sample pocket identification (one-sided kernel)											
dp	-0.74	-0.47	-0.61	-0.32	-0.39	-1.06	-0.59	-1.38 [†]	-0.63	-0.76	0.19
tbl	0.68	0.10	0.37	0.18	0.10	0.23	0.79	1.17	0.62	0.85	0.70
tsp	0.15	-1.06	-1.14	-0.72	-0.85	-0.88	1.49*	2.01**	1.13	1.32*	1.35*
rvar	-1.49 [†]	-0.77	-0.68	-0.64	0.13	-0.20	-1.18	-1.44 [†]	-1.22	-1.67 ^{††}	-1.55 [†]
mv	-0.99	0.03	0.11	0.44	0.45	-0.13	-1.28	-1.54 [†]	-1.06	-1.48 [†]	-1.31 [†]
pc	0.99	0.58	0.28	0.71	-0.19	0.14	0.71	1.39*	0.81	1.53*	1.04
comb1	-0.78	-0.70	-0.67	-0.49	-0.29	-0.62	—	—	—	—	—
comb2	-1.40 [†]	-1.25	-1.06	-0.70	-0.85	-0.88	—	—	—	—	—
comb3	-1.03	-1.20	-1.24	-1.05	-1.21	-1.02	1.07	1.11	0.54	1.00	0.54

Table 5. Replication Results for Out-of-Sample Measures of Prediction Performance with Monthly Data (FST, Table VII, Panel A)

This table compares the prediction performance of models based on the original code from Farmer et al. (2023), which uses a two-sided kernel regression model to identify pockets of predictability (in-sample approach, Panel A) and replication based on a one-sided kernel (out-of-sample approach, Panel B). The presented values are the Clark and West (2007) test statistics (CW) for return predictability benchmarked against a prevailing mean forecast. The estimation period to compute the forecast is 2.5 years, and the calculations assume monthly benchmark specification. The forecast variables include individual predictors: dividend price ratio (dp), Treasury bill rate (tbl), term spread (tsp), and realized variance ($rvar$), as well as composite measures: a recursively computed first principal component of the four individual variables (pc), a four-variable multivariate forecast estimated using a product kernel (mv) and simple averages of univariate forecasts ($comb1$, $comb2$, $comb3$). $comb1$ assigns the forecast from the time-varying coefficient model in-pocket but defaults to the prevailing mean out-of-pocket. $comb2$ takes the average of all models indicating a pocket and the prevailing mean if no model indicates a pocket. $comb3$ consistently utilizes the equal-weighted mean of the four standalone models, regardless of whether in-pocket or out-of-pocket. Finally, pm refers to the prevailing mean forecast. Positive CW values suggest that the model return forecasts outperform the prevalent mean forecast, whereas negative figures signify the opposite. The pockets are defined periods when the fitted squared forecast error differential (SED) calculated using a one-year bandwidth, comparing the prevailing mean with kernel-based forecasts, exceeds zero in the preceding period. We calculate three variants of the forecasts: unrestricted (left section), with the restriction that forecasts of mean excess returns should be non-negative (middle section), and with the sign restriction on slope coefficients (right section). The asterisks *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels from a hypothesis test of $\beta > 0$, and [†], ^{††}, and ^{†††} indicate statistical significance at the 10%, 5%, and 1% levels from a hypothesis test of $\beta < 0$.

Variables	Unrestricted			+ Excess return forecast			All sign restrictions		
	Full sample	In-pocket	Out-of-pocket	Full sample	In-pocket	Out-of-pocket	Full sample	In-pocket	Out-of-pocket
<i>Panel A: In-sample pocket identification (two-sided kernel)</i>									
dp	0.96	4.05***	-0.09	1.03	4.14***	-3.12 ^{†††}	1.13	4.14***	-3.01 ^{†††}
tbl	1.25	3.55***	-0.83	1.23	4.38***	-1.99 ^{††}	2.40***	4.47***	-1.26
tsp	0.78	2.44***	-1.15	0.28	4.75***	-1.45 [†]	0.46	4.93***	-0.20
rvar	0.64	3.28***	0.00	0.40	3.18***	-2.73 ^{†††}	1.04	3.58***	-2.10 ^{††}
mv	1.76**	3.18***	1.28	1.65**	3.70***	-0.69	1.65**	3.70***	-0.69
pc	1.22	3.23***	-1.23	1.04	4.64***	-1.25	1.04	4.64***	-1.25
comb1	4.14***	4.74***	–	4.73***	5.24***	–	4.82***	5.32***	–
comb2	4.48***	5.20***	–	4.81***	5.27***	–	5.48***	6.16***	–
comb3	1.01	2.20**	-2.15 ^{††}	1.10	2.68***	-2.06 ^{††}	1.79**	2.18**	-0.83
<i>Panel B: Out-of-sample pocket identification (one-sided kernel)</i>									
dp	0.96	0.90	0.47	1.03	0.55	0.74	1.13	0.67	0.73
tbl	1.25	1.22	0.41	1.23	1.80**	0.06	2.40***	2.08**	1.53*
tsp	0.78	0.57	0.50	0.28	0.42	0.01	0.46	-0.15	0.51
rvar	0.64	1.01	-0.57	0.40	1.10	-0.29	1.04	1.24	0.58
mv	1.76**	0.57	1.52*	1.65**	0.80	1.49*	1.65**	0.80	1.49*
pc	1.22	1.30*	0.20	1.04	1.88**	-0.49	1.04	1.88**	-0.49
comb1	1.05	1.05	–	1.39*	1.40*	–	1.43*	1.44*	–
comb2	1.04	1.03	–	1.56*	1.58*	–	1.83**	1.84**	–
comb3	1.01	1.22	-0.65	1.10	1.44*	-0.19	1.79**	1.92**	-0.15